CLOUD & DEVOPS

GitOps Visual Guide

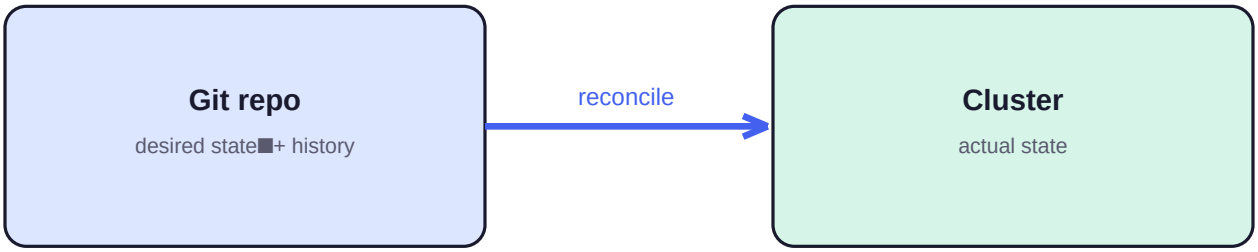
Git as the source of truth —
reconcile your cluster with every commit.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Core Idea: Git Is the Desired State

If it's not in git, it doesn't exist. If it is, the cluster converges to it.



Every change is a pull request: reviewed, tested, auditable — then merged.

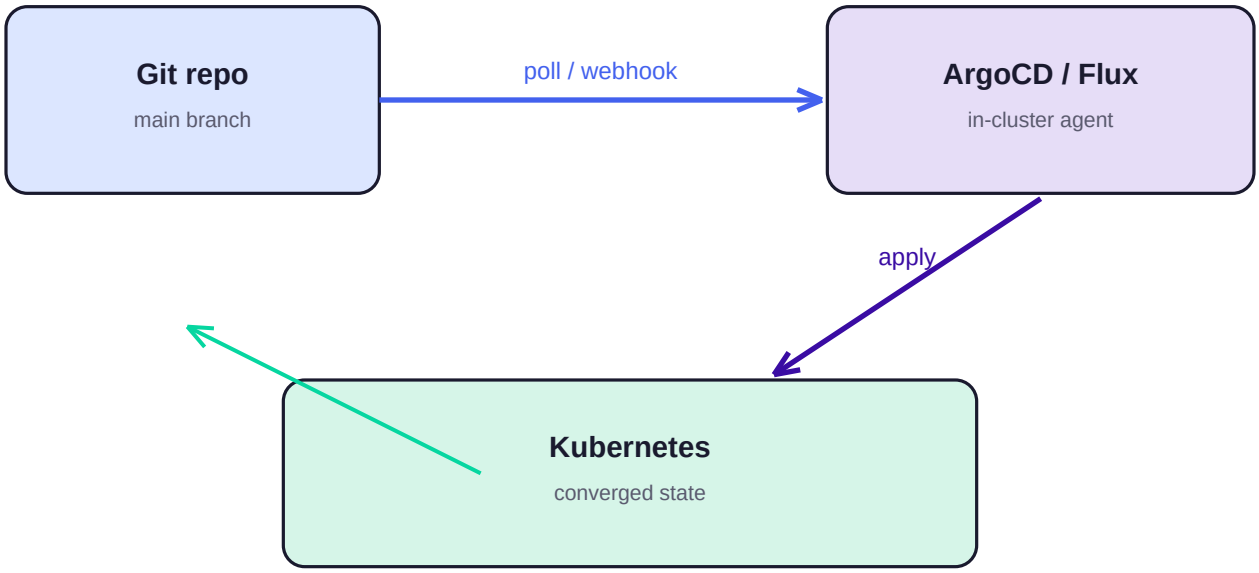
Rollback = git revert. The cluster follows the repo, not the other way around.

Manual `kubectl apply` from a laptop is the anti-pattern — it creates drift nobody can see.

STEP 2/6

The Reconciliation Loop

An agent in the cluster watches git and applies changes continuously.

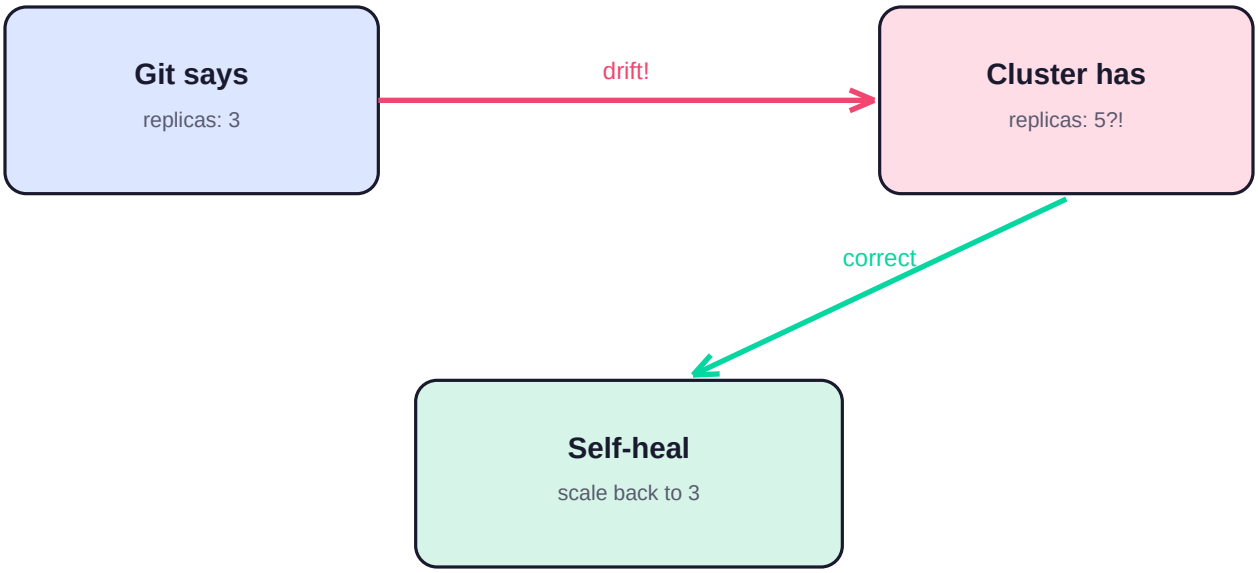


Loop: observe actual state → compare to git → act to close the gap → repeat every few minutes.

STEP 3/6

Drift Detection & Self-Healing

Someone edits the cluster by hand — the agent notices and fixes it.



Self-heal is optional per app — some teams only alert on drift, letting a human decide.

STEP 4/6

Pull vs Push

Two ways to get the desired state into the cluster.

PUSH
CI deploys to cluster

- CI holds cluster credentials (blast radius)
- Drift between pushes goes unnoticed
- Simple for tiny setups

PULL (GitOps)
agent pulls from git

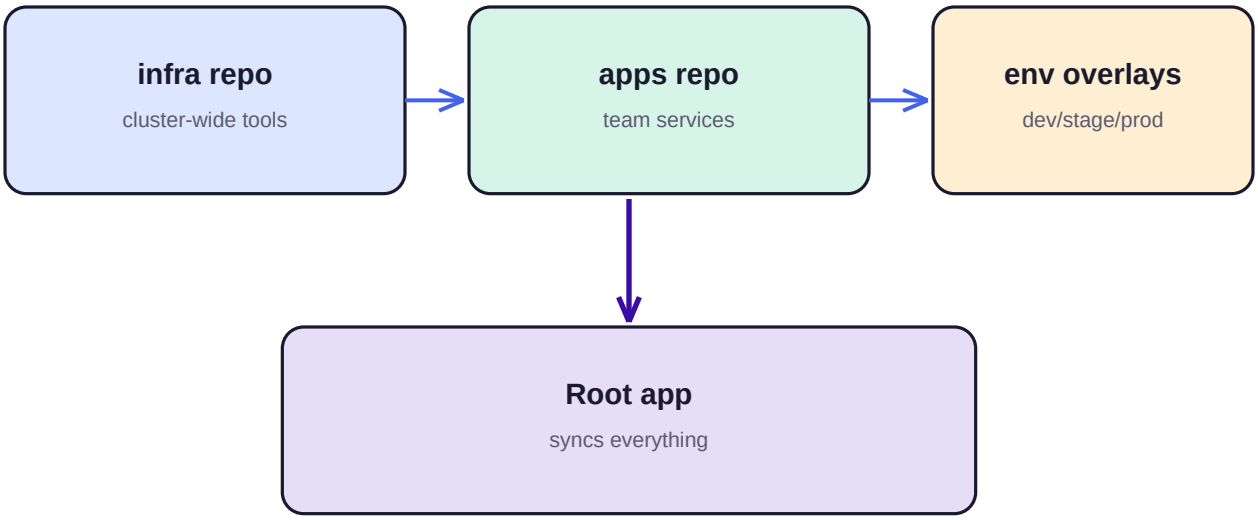
- Cluster needs no inbound creds from CI
- Continuous reconciliation kills drift
- Scales to many clusters from one repo

Pull wins at scale: the cluster's agent is the only writer, and credentials never leave the cluster.

STEP 5/6

Repo Layouts: App of Apps

One repo can bootstrap an entire fleet of clusters.



App-of-apps: one parent application syncs child apps — new cluster = point root app at a new repo path.

STEP 6/6

Secrets in GitOps

Git must never hold plaintext secrets — encrypt or reference them.

Sealed Secrets

encrypted in git, decrypted in-cluster by controller

External Secrets

git holds a reference; operator fetches from vault/KMS

SOPS + age

encrypt values files; CI/operator decrypts at sync

Golden rule: the repo is public-safe. If a secret leaks into git history, rotate it — history is forever.

STEP ✓/✓

Key Takeaways

GitOps — Remember

- Desired state lives in git; the cluster converges to it continuously.
- Reconciliation loop: observe → compare → act, every few minutes.
- Drift is detected automatically; self-heal decides whether to auto-correct.
- Pull beats push at scale: no cluster credentials in CI, drift dies.
- Secrets stay encrypted or referenced — git history never forgets.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev